

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Electrical Engineering) SEMESTER - 6 Summer 2025 (Regular)

Course : Bachelor of Technology (Electrical Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 6

Subject Code & Name: BTEEPE604B - SMART GRID TECHNOLOGY

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

- 1 What does AMR stand for?
 - a) Advanced Metering Report
 - b) Automatic Metering Recalculation
 - c) Automatic Meter Reading
 - d) Automated Metering Relay
- GPS is used in Smart Grid for:
 - a) Weather forecasting
 - b) Time synchronization
 - c) Entertainment
 - d) Advertising
- Power Line Communication (PLC) uses:
 - a) Existing electrical wiring for data
 - b) Fiber cables only
 - c) Satellite link
 - d) Water pipelines
- 4 Which of the following is a computational intelligence technique?
 - a) Fuzzy logic
 - b) Traditional relay
 - c) Oil circuit breaker
 - d) Overhead insulation
- 5 Plug-in hybrids are:
 - a) Manual vehicles
 - b) Vehicles using both battery and fuel engine
 - c) Water-powered cars
 - d) Cable-controlled cars
- 6 Which is a type of cyber attack on Smart Grid?
 - a) Fake Data Injection
 - b) Line shorting
 - c) Overheating
 - d) Magnet tampering
- 7 Voltage control in Microgrid ensures:
 - a) Overvoltage
 - b) Stable voltage supply
 - c) Line drop
 - d) No power

- 8 What is PMU?
a) Power Monitoring Unit
b) Phasor Measurement Unit
c) Passive Measuring Unit
d) Phase Mode Unit
- 9 What is the environmental benefit of Smart Grids?
a) Increase in pollution
b) Deforestation
c) Reduction in emissions
d) Increase in noise
- 10 Which technology is commonly used in Smart Grid communication?
a) Cable TV b) ZigBee c) Remote control d) VCR
- 11 Artificial Intelligence in Smart Grid is used for:
a) Manual calculations
b) Load forecasting and optimization
c) Increasing manual errors
d) Reducing automation
- 12 What is SMES in Smart Storage?
a) Smart Metering Energy System
b) Static Metering Energy System
c) Superconducting Magnetic Energy Storage
d) Smart Magnetic Energy Storage
- Q2. Solve the following.
- What are the key challenges faced by Indian Smart Grid implementation? 6
- Write a short note on Renewable Integration in Smart Grid architecture. 6
- Q3. Solve the following.
- A) Discuss the impact of Smart Grid on air pollutant emissions and reliability of power supply. 6
- B) Draw and explain a typical architecture of Smart Grid with its major components. 6
- Q4. Solve Any Two of the following.
- A) What is Distributed Generation (DG)? Explain with advantages and limitations. 6
- Discuss the role of Electric Vehicles (EVs) and plug-in hybrids in Smart Grid. 6
- Describe the challenges and opportunities of integrating Renewable Energy Technologies in Smart Grids. 6
- Q5. Solve Any Two of the following.
- What is Advanced Metering Infrastructure (AMI)? Explain its components. 6
- B) Explain Load Frequency Control (LFC) in a Microgrid system. 6
- C) What is Power Line Communication (PLC) and how is it used in Smart Grid? 6
- Q6. Solve Any Two of the following.
- A) Discuss cyber security challenges in Smart Grid and their potential consequences. 6
- B) What are Load Altering Attacks and False Data Injection Attacks in Smart Grids? 6
- C) Write a short note on any two: Bluetooth, ZigBee, GPS, Wi-Fi and Wi-Max. 6